

Hybrid Hidden Markov Model for Modeling Equity Excess Growth Rate Dynamics: A Discrete-State Approach with Jump-Diffusion

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Abstract

1 Financial time series exhibit statistical regularities, commonly referred to as
2 stylized facts, that are not captured by classical Gaussian diffusion models.
3 In particular, empirical equity excess growth rates display heavy-tailed dis-
4 tributions, negligible linear autocorrelation, and volatility clustering. This
5 work proposes a hybrid hidden Markov framework that discretizes continu-
6 ous excess growth rates into quantile-defined market states and augments the
7 regime-switching process with a Poisson-driven jump-diffusion mechanism.
8 Model parameters are estimated empirically through transition counting,
9 avoiding the computational cost of likelihood maximization and expectation-
10 maximization procedures. A pipeline was developed to support the construc-
11 tion of excess growth rate models for any ticker; we validated the framework
12 using SPY (Standard & Poor’s 500 index exchange-traded fund) as an ex-
13 ample asset, though the framework is applicable to any ticker within our
14 424-asset dataset. Using 10-years of trading data, we show that while stan-
15 dard regime-switching models reproduced marginal excess growth rate dis-
16 tributions, the jump-duration mechanism was needed to generate persistent
17 high-volatility regimes. The selection of the number of states provided a bal-
18 ance between distributional fidelity and statistical robustness, reproducing
19 heavy-tailed behavior and volatility clustering in both in-sample and out-of-
20 sample settings. Taken together, this hybrid hidden Markov model offers a
21 generative framework for simulating market dynamics and producing syn-
22 thetic time series that remain statistically consistent with underlying market
23 regimes.

Keywords: Portfolio Optimization, Cobb-Douglas Utility Function, Single

24 Introduction

25 The quantitative modeling of financial markets has long faced the chal-
26 lenge of reconciling the theoretical implications of the efficient markets hy-
27 pothesis (EMH) with the empirical complexity of observed price dynamics.
28 Under the EMH, asset prices fully reflect all available information, implying
29 that price changes follow a martingale process with no exploitable predictabil-
30 ity [1]. This perspective motivated the widespread adoption of geometric
31 Brownian motion (GBM) as a canonical model for asset prices, yet decades
32 of data show systematic departures from Gaussian assumptions, including
33 heavy-tailed excess growth rate distributions, volatility clustering, and in-
34 termittent extreme events [2, 3]. These stylized facts are observed across
35 asset classes and time periods, and they matter because they govern tail
36 risk, drawdown behavior, and the persistence of stress regimes that shape
37 portfolio outcomes.

38 This study focuses on three core properties: leptokurtic excess growth
39 rate distributions, the absence of linear autocorrelation in raw excess growth
40 rates, and the persistence of volatility. The first two properties motivate
41 models that can preserve marginal distributional shape without introducing
42 spurious predictability, while the third points to slow-moving higher-order
43 dependence that is not captured by diffusion alone. A range of alternative
44 approaches address these issues from different angles: autoregressive condi-
45 tional heteroscedasticity (ARCH) and generalized autoregressive conditional
46 heteroscedasticity (GARCH) models target conditional variance dynamics
47 [4, 5]; stochastic-volatility and jump-diffusion models introduce latent vari-
48 ance or discontinuities to enrich tails [6, 7]; and data-driven sequence mod-
49 els, including recurrent neural networks (RNNs) and long short-term mem-
50 ory (LSTM) variants, have been used for excess growth rate and volatility
51 prediction and, in probabilistic form, can emit full predictive distributions
52 [8, 9, 10, 11, 12, 13, 14]. While useful, these approaches either empha-
53 size conditional variance over regime duration, require intensive likelihood-
54 based estimation, or sacrifice interpretability and explicit control over regime
55 persistence. Hidden Markov models (HMMs) and related regime-switching
56 frameworks offer a probabilistic structure for non-stationary market behavior
57 through latent state dynamics [15], yet standard HMM specifications often

fail to generate persistent high-volatility regimes, leading to overly rapid reversion following extreme events [16, 17]. Bridging this gap requires a model that preserves distributional fidelity while also enforcing realistic duration in extreme regimes.

We address this challenge by developing a hybrid HMM framework that discretizes continuous excess growth rates into quantile-based regimes and augments the resulting Markov process with a jump-duration mechanism that enforces volatility persistence, linking smooth diffusion to discontinuous shocks identified by jump-detection methods [18]. Using a 424-asset dataset and SPY as a case study, we estimate transition dynamics empirically and validate the model through in-sample and out-of-sample tests on distributional fit and volatility clustering. We find that a state resolution of $N = 90$ provides a balance between fidelity and statistical robustness, and that the jump-duration extension is needed for reproducing persistent high-volatility regimes that standard regime-switching models miss. Future work will extend the framework to multi-asset portfolios, alternative state encodings, and higher-frequency data where jump identification and regime persistence play a larger role in stress testing and scenario design.

Related Work

77 **Methods**

78 *Data.*

79 The empirical foundation of this study was a dataset composed of 424
80 United States-listed equities and exchange-traded funds spanning 10 years
81 (2014-2024). A pipeline was developed to support the automated construc-
82 tion of excess growth rate models for any ticker within this dataset, allowing
83 for batch simulation and stress testing across market sectors. To validate
84 the performance of this pipeline, we focused our analysis on the single-asset
85 SPY, which tracked the Standard & Poor’s (S&P) 500 index. The data in-
86 cluded daily open, high, low, and close prices along with volume-weighted
87 average price metrics (VWAP). For training purposes, the first 2,766 obser-
88 vations constituted the in-sample dataset, spanning from January 3, 2014,
89 to December 31, 2024. A subsequent 220 trading days, from January 3, to
90 November 18, 2025, were reserved for out-of-sample testing to assess model
91 generalizability.

92 *Excess growth rate calculation.*

93 We computed the excess growth rate $G_{i,j}$ for ticker i between time pe-
94 riod $j - 1 \rightarrow j$ for a given equity price series $P_{i,j}$ (units: dollars per share
95 (USD/share)). To isolate the stochastic premium earned over a risk-free
96 asset, we computed the excess growth rate for ticker i during the period
97 $j - 1 \rightarrow j$ as:

$$G_{i,j} \equiv \left(\frac{1}{\Delta t} \right) \cdot \ln \left(\frac{P_{i,j}}{P_{i,j-1}} \right) - r_f \quad (1)$$

98 where Δt represented the time step set to 1/252 for daily data (units: years),
99 and r_f denotes the constant continuously compounded risk-free rate (units:
100 year⁻¹) derived from Separate Trading of Registered Interest and Principal
101 of Securities (STRIPS) bond yields observed on September 10, 2025. Un-
102 like return, the excess growth rate has units of inverse time (year⁻¹) and
103 directly quantifies the premium earned above a continuously compounded
104 risk-free benchmark. Log growth rates are time-additive and consistent with
105 a continuous compounding framework. While r_f varied dynamically over the
106 sample period, a constant proxy was employed here to isolate the endogenous
107 volatility dynamics without introducing exogenous rate-curve noise.

108 *Discrete hidden Markov model.*

109 We defined the hidden Markov model using the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$
 110 [15]. The hidden state $S_t \in \mathcal{S} = \{1, \dots, N\}$ represented market mood
 111 regimes and was defined by quantile bins of a fitted cumulative distribu-
 112 tion function (CDF) for the excess growth rate series. The observation space
 113 $\mathcal{O} \subset \mathbb{R}$ consisted of continuous excess growth rate values, the transition
 114 matrix $\mathbf{T}$ governed regime-to-regime dynamics, the emission model $\mathbf{E}$ was
 115 parameterized by state-conditional Normal distributions, and $\bar{\pi}$ denoted the
 116 stationary distribution. The observation model was

$$G_t \mid S_t = k \sim \mathcal{N}(\mu_k, \sigma_k). \quad (2)$$

117 To map observations to hidden states during estimation, we defined quantile
 118 boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \dots, N - 1\}$ and used finite
 119 outer bounds $Q_0 = F_L^{-1}(0.001; \mu_L, b_L)$ and $Q_N = F_L^{-1}(0.999; \mu_L, b_L)$. An
 120 observation was assigned to state k when $Q_{k-1} < R_t \leq Q_k$.

121 The choice of the Laplace distribution over alternatives such as the Stu-
 122 dent’s t-distribution was motivated by its alignment with the peaks and
 123 heavy tails of high-frequency financial excess growth rates [3, 19]. While the
 124 Student’s t-distribution was often used to model leptokurtosis, the Laplace
 125 distribution’s peak at the mean matched the concentration of small price
 126 movements observed in the SPY dataset [2, 20]. Furthermore, the Laplace
 127 distribution provided a closed-form for quantile estimation, supporting the
 128 424-ticker pipeline. While maximum likelihood estimation (MLE) for the
 129 Student’s t-distribution often required iterative numerical optimization, the
 130 Laplace distribution allowed for a direct MLE approach [21]. The quantile
 131 boundaries were derived from a parametric Laplace fit and therefore did not
 132 enforce equal counts per bin in the historical data; instead, they provided a
 133 partition of the distribution that preserved tail shape. By using the Laplace
 134 distribution to inform the binning process, the model used this baseline to
 135 characterize the peak and tails of the excess growth rate series before the
 136 empirical transition logic and jump-diffusion mechanism were applied [22].

137 We estimated the transition matrix $\mathbf{T}$ through direct frequentist count-
 138 ing rather than iterative optimization. This approach differed from the tra-
 139 ditional expectation-maximization (EM) algorithm used for parameter esti-
 140 mation in Gaussian mixture and hidden Markov models [21]. However, this
 141 approach was justified because the discrete state assignments allowed for
 142 straightforward counting of observed transitions between regimes, and it was
 143 conceptually simple to implement and understand.

Empirical transition matrices often had low probabilities for exiting central states into tail states, which caused models to lose the effect of volatility clustering too quickly [17]. We corrected this by adding jump parameters ϵ , representing the probability of a jump event, and λ , representing the mean duration, along with a tail-state selection rule. Tail-states were defined as $\mathcal{S}_{bottom} = \{1, \dots, N_{tail}\}$ and $\mathcal{S}_{top} = \{N - N_{tail} + 1, \dots, N\}$, where N_{tail} was user-configurable. If a jump was triggered, the jump-duration K was sampled from a Poisson distribution such that $K \sim \text{Poisson}(\lambda)$, which was standard for modeling independent events in fixed intervals [23]. During these K steps, the standard Markovian transitions were overridden to target tail-states, with a user-configurable bias p_{neg} (default 0.52) that favored negative-tail states to reproduce gain/loss asymmetry.

To characterize the excess growth rates within each regime, we parameterized the emission model with Normal distributions $\mathcal{N}(\mu_k, \sigma_k)$ whose bounds coincided with the 0.1 and 99.9 percentiles of the state-conditional distribution. Specifically, we set $\mu_k = (Q_{k-1} + Q_k)/2$ and $\sigma_k = (Q_k - Q_{k-1})/(2z_{0.999})$, where $z_{0.999} \approx 3.09$ was the standard normal 99.9th percentile. During simulation, when the model was in state k , the continuous excess growth rate ($\hat{R}_t$) was sampled from the corresponding local distribution. This hybrid approach ensured that the model preserved the discontinuous components of the price process important for asset modeling while maintaining a smooth representation of intra-state variance [22].

Growth rate generation.

Let's outline a simple sampling algorithm for generating random sequences from an HMM (without jumps). We initialize the generation process by drawing an initial hidden state X_0 from the stationary distribution $\bar{\pi}$. The stationary distribution $\bar{\pi}$ represented the long-run probability of the market occupying various regimes or moods [24]. While $\bar{\pi}$ mathematically satisfies the balance equation $\bar{\pi} = \bar{\pi}\mathbf{T}$, we estimated this distribution numerically by raising the transition matrix $\mathbf{T}^n$ to a sufficiently large power $n = 50$, ensuring the vector reflected the equilibrium of market dynamics [25].

Given an HMM defined by the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{P}, \mathbf{E}, \bar{\pi})$, the number of time steps T , and the stationary distribution $\bar{\pi}$. Draw an initial hidden state X_0 from the initial distribution: $X_0 \sim \bar{\pi}$. For $n = 1$ to T , we sample the next hidden state X_n from the `categorical` distribution $\mathbb{P}_H(\dots)$ defined by the previous state X_{n-1} and the transition matrix $\mathbf{T}$: $X_n \sim \mathbb{P}_H(X_n | X_{n-1})$. Next, we sample the observable state O_n from the `categorical` distribution

181 $\mathbb{P}_E(\dots)$ defined by the current hidden state X_n and the emission matrix $\mathbf{E}$:
 182 $O_n \sim \mathbb{P}_E(O_n \mid X_n)$. Finally, we store the sequences of hidden states and
 183 observable states. This procedure generates a sequences of hidden states
 184 $\{X_1, X_2, \dots, X_T\}$ and observable states $\{O_1, O_2, \dots, O_T\}$. The initial state
 185 X_0 is used to generate the sequence but is not included in the output.

186 *Hyperparameter optimization via grid search.*

187 To rigorously identify the optimal jump probability (ϵ) and mean jump
 188 duration (λ) [7, 22], we implemented a multi-objective grid search designed
 189 to minimize the discrepancy between the historical and simulated market sig-
 190 natures. The error function specifically targeted two stylized facts prevalent
 191 in financial time series [3, 16]: volatility clustering and leptokurtosis. We
 192 measured volatility clustering using the sum of squared errors between the
 193 observed and simulated autocorrelation function (ACF) of absolute excess
 194 growth rates [26] up to a lag of $L = 252$ trading days (approximately one
 195 trading year). To preserve the heavy-tailed nature of the distribution ini-
 196 tially noted in foundational market studies [2], we included a penalty term
 197 for the global kurtosis.

198 The objective function $J(\epsilon, \lambda)$ was formalized as:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^L (\text{ACF}_{obs}(\tau) - \overline{\text{ACF}}_{sim}(\tau))^2 + w_K (K_{obs} - \overline{K}_{sim})^2 \quad (3)$$

199 where $\text{ACF}_{obs}(\tau)$ and K_{obs} represented the empirical absolute growth au-
 200 tocorrelation at lag τ and the global kurtosis, respectively. The simulated
 201 equivalents, $\overline{\text{ACF}}_{sim}(\tau)$ and $\overline{K}_{sim}$, were averaged across 1,000 independent
 202 synthetic paths of 2,766 trading days to ensure statistical stability [23]. A
 203 penalty weight of $w_K = 0.20$ was assigned to prioritize tail behavior alongside
 204 the temporal ACF decay. The grid search systematically explored ϵ values
 205 from 0.00001 to 0.0001 and λ values from 1.0 to 100.0. The full computational
 206 procedure is detailed in Algorithm 4.

207 *Statistical validation metrics*

208 *Sensitivity analysis: low-resolution models*

209 This section provides the results for lower state resolutions ($N = 30$ and
 210 $N = 60$) to demonstrate how state granularity impacts the model’s ability to
 211 capture stylized facts. Table 2 summarizes the pass rates for the Kolmogorov-
 212 Smirnov and Anderson-Darling tests across these four cases. As the number

²¹³ of states decreases, the model maintained overall distribution fit, though tail-
²¹⁴ region precision decreased slightly compared to the 90-state configuration.

215 Empirical Study

216 We simulated 1,000 paths of 2766 trading days using VWAP price data for
 217 both the standard model with no-jumps and the hybrid model with jumps.
 218 Applying the multi-objective grid search detailed in the Methods, the optimal
 219 jump hyperparameters were identified as ($\epsilon \approx 5 \times 10^{-5}$ and $\lambda \approx 67$). This
 220 configuration successfully reproduced the heavy-tailed kurtosis and ARCH
 221 effect observed in SPY (Figure 16). Based on the statistical consistency
 222 results, the configuration with a heuristic of $90 \leq N \leq 100$ states was used
 223 in this study. Increasing the number of states N increased the granularity
 224 of the market regimes, leading to a closer fit for in-sample results. However,
 225 there is a limit to this refinement; if N was too large, such as 350 states, the
 226 empirical estimation protocol failed because the historical dataset became
 227 too sparse relative to the state space. In such cases, certain states could
 228 remain unvisited, resulting in zero transition probabilities that broke the
 229 mathematical integrity of the Markov chain. Conversely, a very small number
 230 of states, such as $N = 10$, resulted in a coarse model that missed the heavy
 231 tails observed in the empirical data. Therefore, the selection of $N = 90$
 232 provided a balance between distributional fidelity and statistical robustness.

233 We performed Kolmogorov-Smirnov (KS) and Anderson-Darling (AD)
 234 tests to characterize the statistical consistency of the simulated excess growth
 235 rate distributions against the historical data. Both the KS and AD tests are
 236 non-parametric goodness-of-fit tests that compare the empirical cumulative
 237 distribution functions (CDFs) of two datasets [27, 28, 29]. The KS test
 238 focused on the maximum distance between the two CDFs, while the AD test
 239 placed more weight on the tails of the distributions.

240 In-sample KS and AD tests showed pass rates above 98% for both model
 241 variants (Table 1), indicating distributional alignment across the 10-years of
 242 training data. The standard no-jumps model matched the marginal distribu-
 243 tion but failed to capture volatility clustering (Figure 5). The autocorrelation
 244 of absolute growth in subplot (c) remained flat, showing that state resolution
 245 alone could not overcome the memoryless limitations of standard transition
 246 logic. The hybrid model reproduced volatility clustering, with the absolute-
 247 growth autocorrelation mirroring the ARCH effect (Figure 16). With $N = 90$
 248 states, this configuration provided a balance and aligned with volatility esti-
 249 mation techniques using HMM architectures [30].

250 Out-of-sample simulations remained statistically consistent, with KS and
 251 AD pass rates of 83 percent and 80 percent, respectively (Figure 11). Fill

me in. Out-of-sample volatility differed by approximately 32 percent, yet volatility clustering persisted in subplot (c) (Figure 11). Unlike the standard Markov transition matrix, which decayed rapidly, the Poisson jump mechanism mimicked the persistence of market volatility in unseen conditions. These results validated the hybrid framework as a tool for producing synthetic time series that remained statistically consistent with the underlying stochastic process [3, 16, 17].

The correlation-distance distribution was centered at a mean distance of 1.0, indicating temporal independence between simulated paths and the historical trajectory (Figure 6). The correlation distance, defined as $(1 - \rho)$, measured the temporal alignment between simulated paths and the observed historical trajectory. This result confirmed that the generative framework produced independent stochastic realizations that captured the stylized facts of the market without violating the efficient markets hypothesis by attempting to predict specific events [1]. This underscored the distinction between a generative framework intended for stress testing and a forecasting tool intended for point prediction [31, 32].

269 Discussion

270 This study introduced a hybrid hidden Markov modeling framework to
 271 replicate the statistical regularities of empirical equity excess growth rates.
 272 By integrating a quantile-discretized state space with a Poisson-driven jump-
 273 duration mechanism, we developed a generative framework that captured
 274 the leptokurtic distributions and volatility clustering observed in the SPY
 275 exchange-traded fund. Our results confirmed that while increasing the state
 276 resolution to $N = 90$ provided the granularity to characterize the central
 277 mass of the excess growth rate distribution, standard Markovian transitions
 278 alone did not reproduce higher-order temporal dependencies. The inclusion
 279 of discontinuous shocks through a jump-diffusion extension was needed to
 280 drive the model into persistent high-volatility regimes, thereby replicating
 281 the ARCH effect identified in historical data.

282 Statistical validation using Kolmogorov-Smirnov and Anderson-Darling
 283 tests demonstrated high in-sample consistency, with pass rates exceeding 98
 284 percent. Although the out-of-sample performance showed an expected de-
 285 crease in fidelity, the hybrid model maintained a baseline with pass rates
 286 of 83 percent and 80 percent, respectively. Furthermore, the correlation
 287 distance analysis confirmed that our framework produced independent real-
 288 izations that remained uncorrelated with specific historical timing; this sat-
 289 isfied the requirements of the efficient markets hypothesis and distinguished
 290 our generative methodology from predictive forecasting tools [1, 31, 32]. The
 291 proposed framework offered an alternative to standard likelihood-based esti-
 292 mation methods by relying on empirical estimation. By bypassing the com-
 293 plexities of the expectation-maximization algorithm through direct frequen-
 294 tist counting, the model remained suited for large-scale stress testing and
 295 risk management applications [33, 34, 23]. Future research could extend this
 296 discrete-state logic to multi-asset portfolios or incorporate foundation mod-
 297 els for inference to further refine the detection of Markov jump processes
 298 in higher-frequency data [35, 36]. Ultimately, the hybrid hidden Markov
 299 model served as a foundation for simulating market dynamics and generating
 300 synthetic time series that remained statistically consistent with underlying
 301 market regimes [37, 24].

302 While the current framework was able to generate statistically consistent
 303 excess growth rate distributions, there were several limitations that could
 304 be addressed in future work. First, the model assumed stationarity in tran-
 305 sition dynamics, which may not hold during periods of structural market

306 change. Future extensions could incorporate time-varying transition matrices
307 or regime-dependent jump parameters to capture evolving market conditions.
308 Second, the use of a fixed quantile-based state space may limit the model's
309 ability to adapt to changing volatility regimes; adaptive binning strategies
310 or non-parametric state definitions could enhance flexibility. Third, while
311 the Laplace distribution provided a good fit for excess growth rates, alter-
312 native distributions or mixture models could be explored to better capture
313 tail behavior and skewness. Finally, the current implementation focused on
314 univariate excess growth rates; extending the framework to multivariate set-
315 tings could enable portfolio-level synthetic allocation studies, stress testing
316 and risk assessment.

317 **Conclusion**

318 **Conflict of Interest Statement**

319 The authors declare that the research was conducted without any com-
320 mercial or financial relationships that could potentially create a conflict of
321 interest.

322 **Author Contributions**

323 J.V. directed the study. A.A. developed the model and simulation code,
324 conducted the in-sample and out-of-sample analysis, and generated the fig-
325 ures. Both authors edited and reviewed the final manuscript.

326 **Data Availability Statement**

327 The model code, simulation scripts, training and testing data are avail-
328 able under a Massachusetts Institute of Technology (MIT) license from the
329 GitHub repository: <https://github.com/varnerlab/HMM-Model-with-Jumps>.
330 git

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0.48

(a)

b

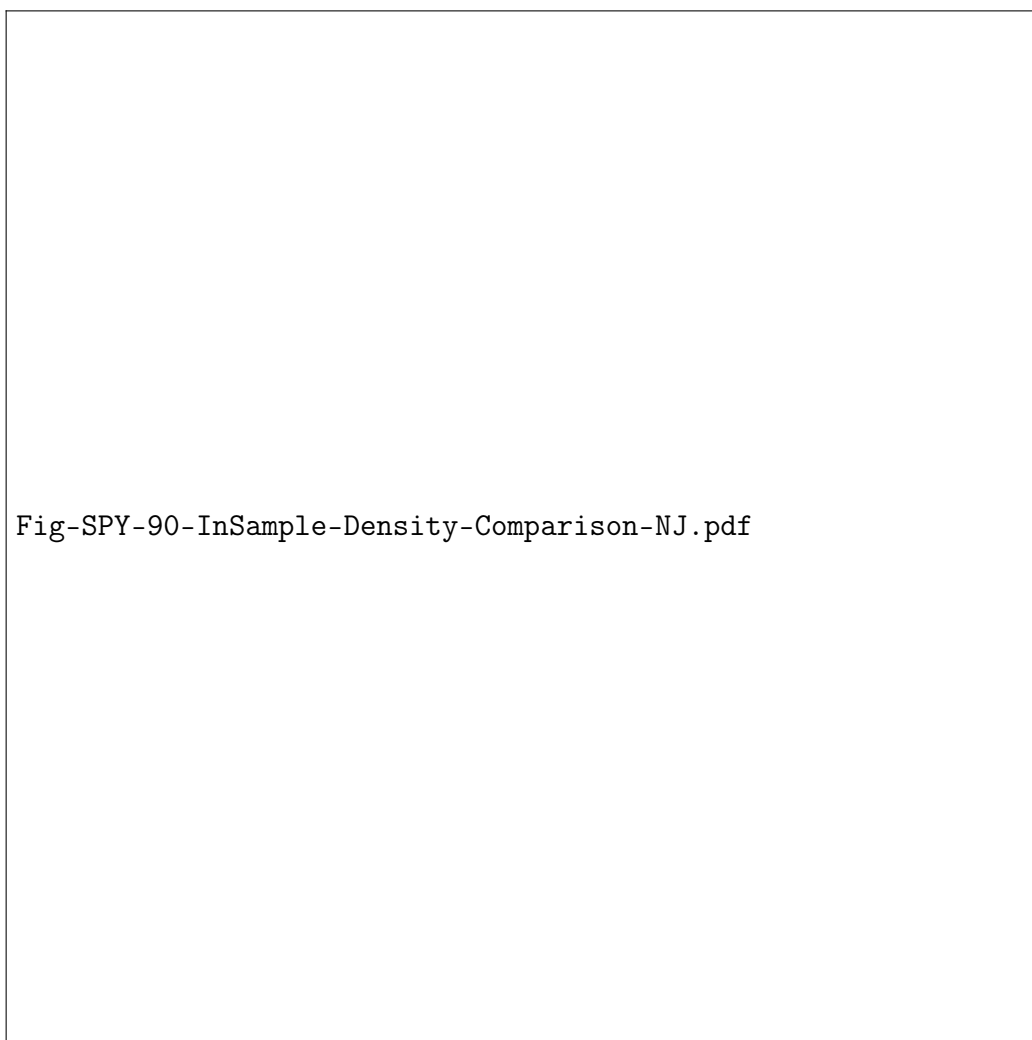
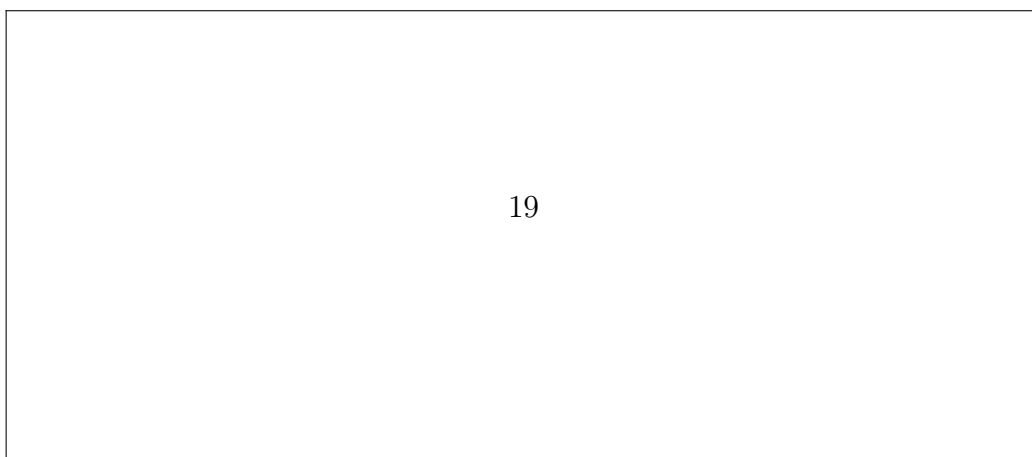


Figure 1: Probability density comparison

0.48

(a)

b



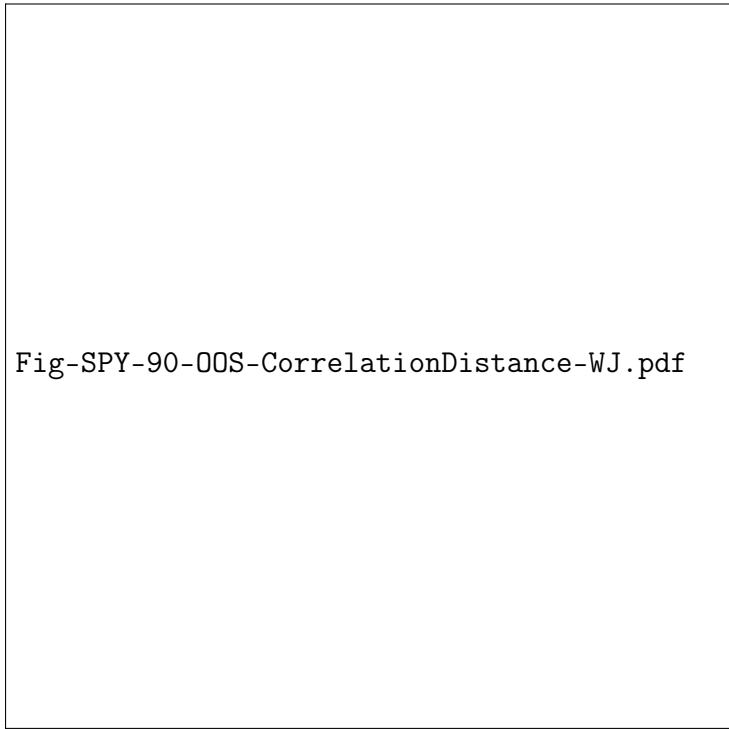


Fig-SPY-90-OOS-CorrelationDistance-WJ.pdf

Figure 6: Distribution of correlation distances for 1,000 simulated paths. The mean distance of 1.0 for the correlation distance $(1 - \rho)$ confirmed the generation of independent realizations consistent with the efficient markets hypothesis.

0.48

(a)

b

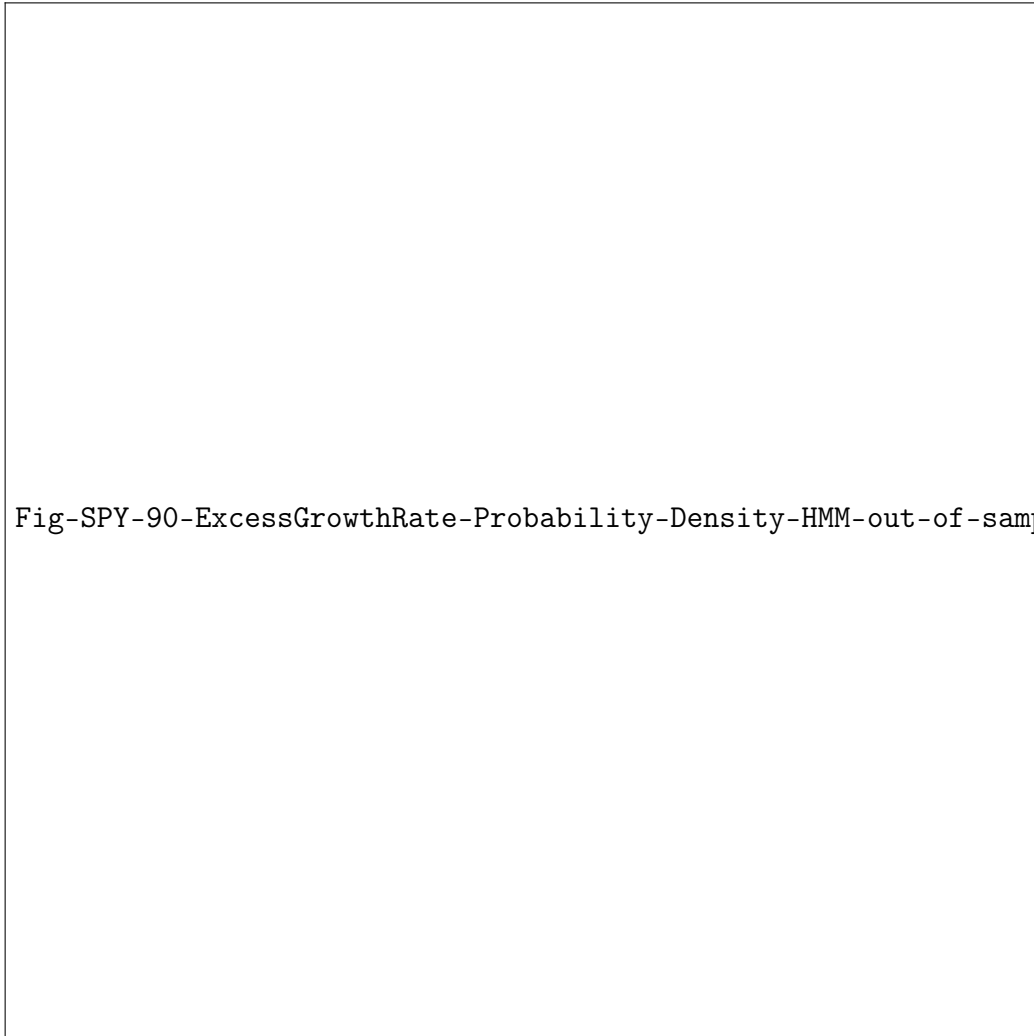


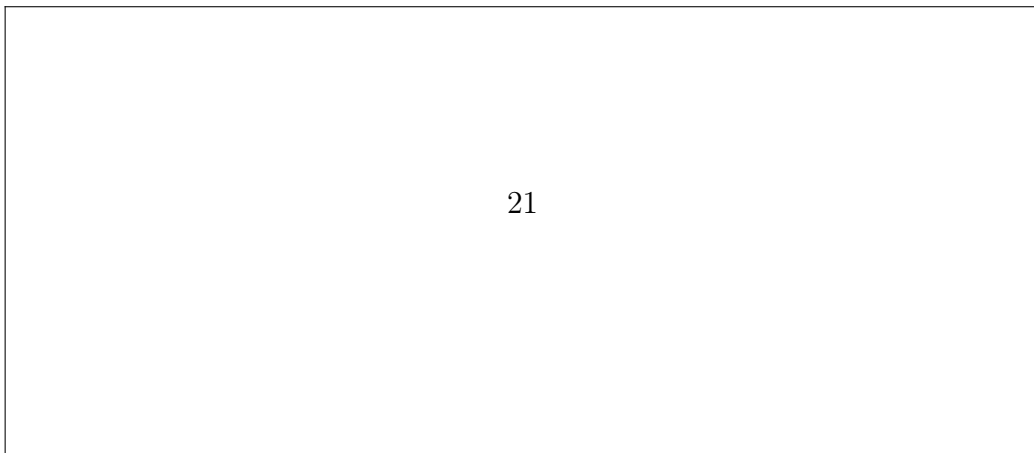
Fig-SPY-90-ExcessGrowthRate-Probability-Density-HMM-out-of-sample.pdf

Figure 7: Probability density comparison

0.48

(a)

b



0.48

(a)

b

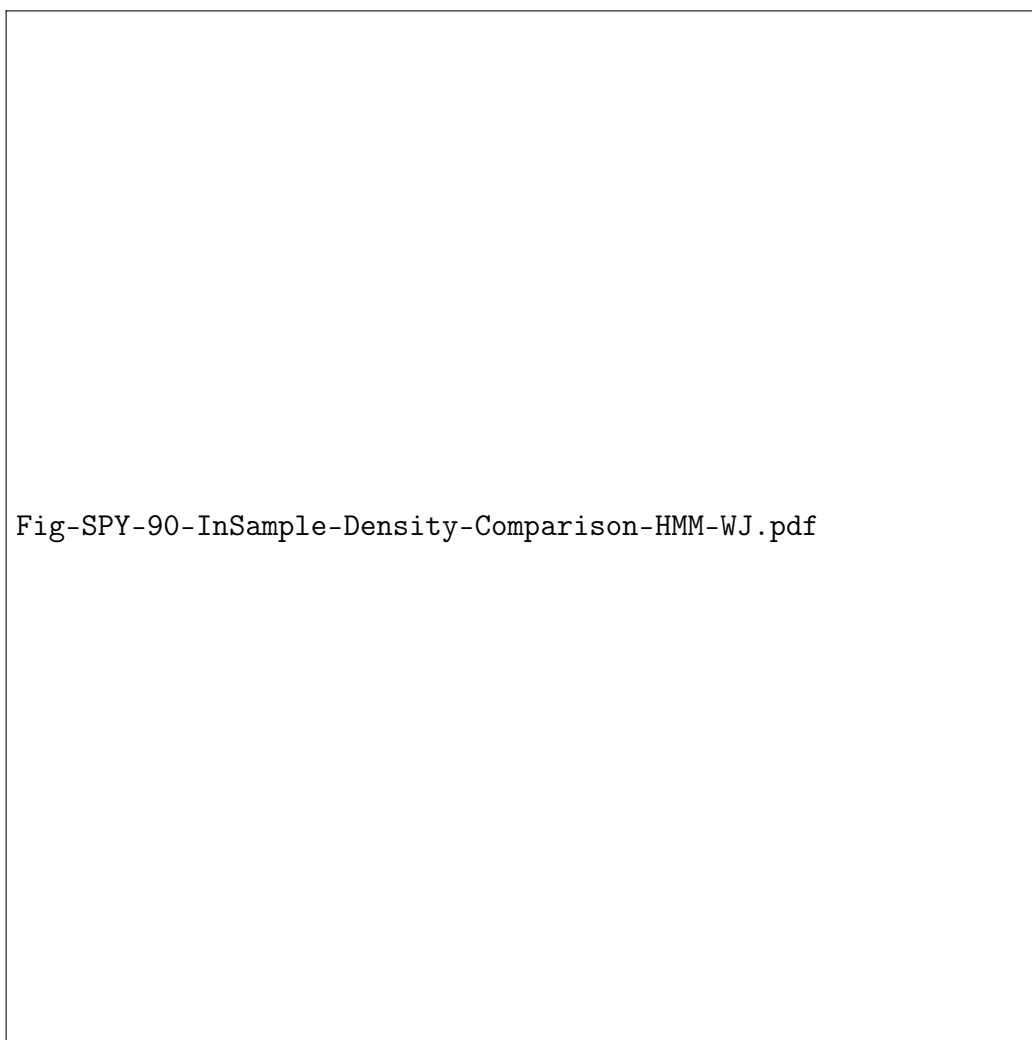


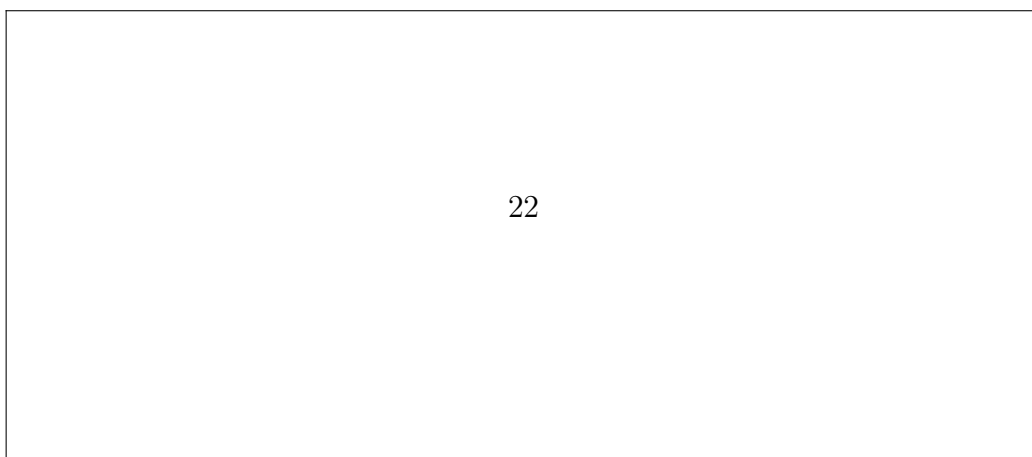
Fig-SPY-90-InSample-Density-Comparison-HMM-WJ.pdf

Figure 12: Probability density comparison

0.48

(a)

b



The following figures illustrate the performance of these low-resolution configurations. Figure 19 and Figure 22 show the results for 30 states, while Figure 25 and Figure 28 provide the comparison for 60 states.

Table 1: Statistical consistency of simulated excess growth rates across hidden Markov model configurations.

model type	number of states (N)	KS test pass rate	AD test pass rate
standard (no-jumps)	90	99.4%	98.4%
hybrid (with jumps)	90	99.5%	98.1%

Table 2: Statistical consistency of simulated growth rates for lower state resolutions ($N = 30$ and $N = 60$) using a p -value cutoff of 0.05.

model type	number of states (N)	KS test pass rate	AD test pass rate
standard (no-jumps)	60	99.4%	98.3%
standard (no-jumps)	30	99.2%	98.2%
hybrid (with jumps)	60	99.3%	97.6%
hybrid (with jumps)	30	98.7%	97.7%

Algorithms

0.48

(a)

b

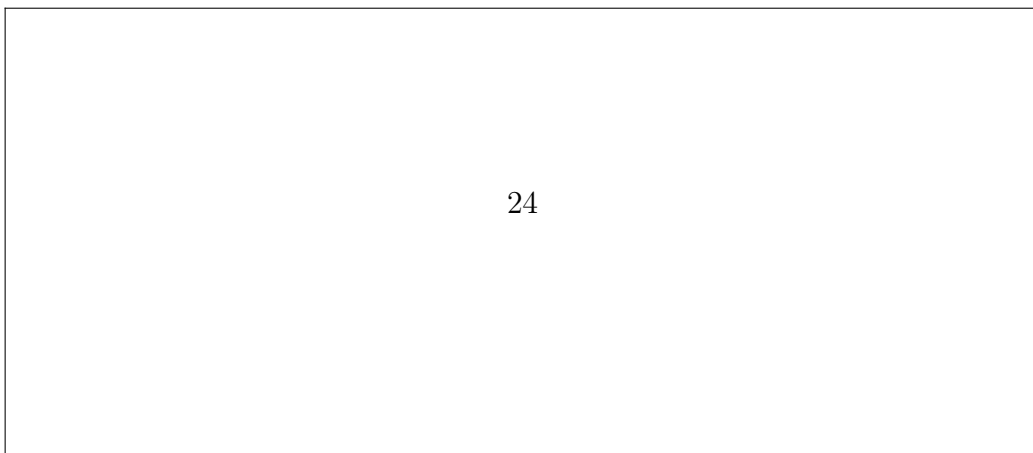


Figure 17: Density (standard)

0.48

(a)

b



0.48

(a)

b

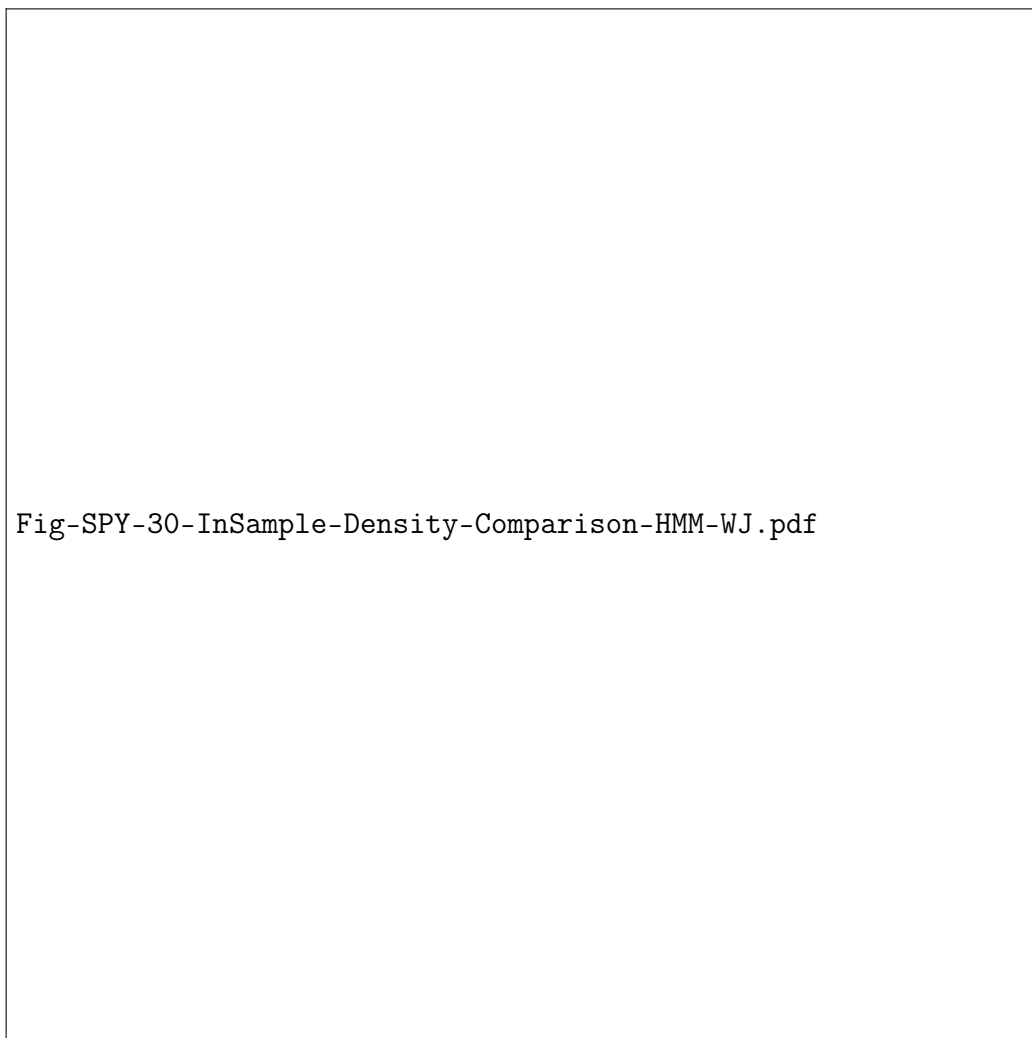


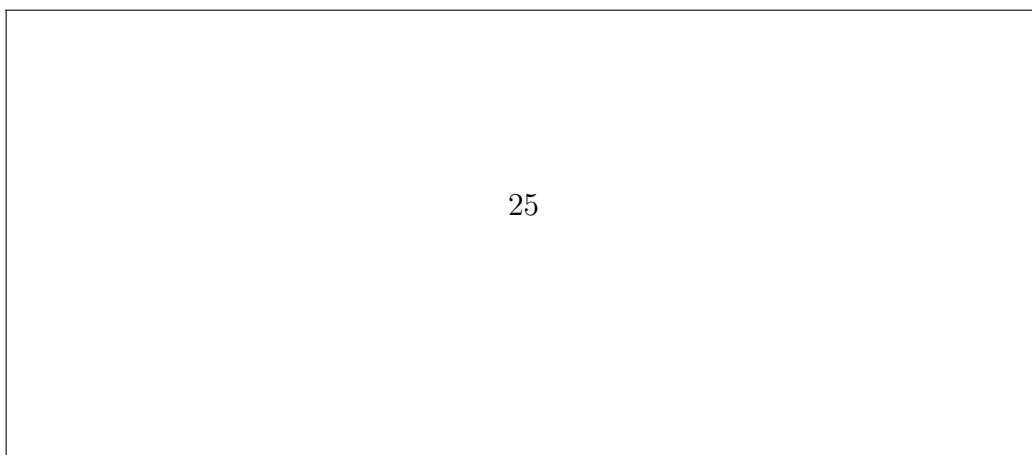
Fig-SPY-30-InSample-Density-Comparison-HMM-WJ.pdf

Figure 20: Density (hybrid)

0.48

(a)

b



0.48

(a)

b

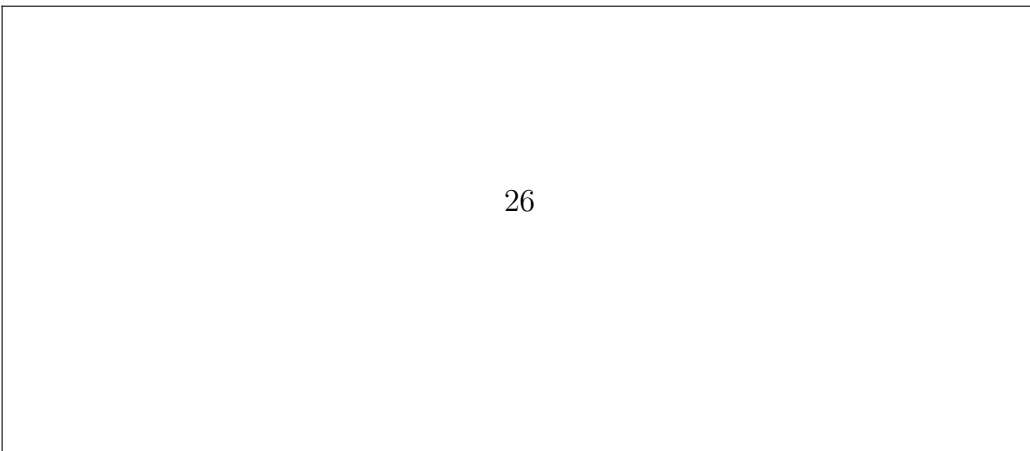


Figure 23: Density (standard)

0.48

(a)

b



0.48

(a)

b

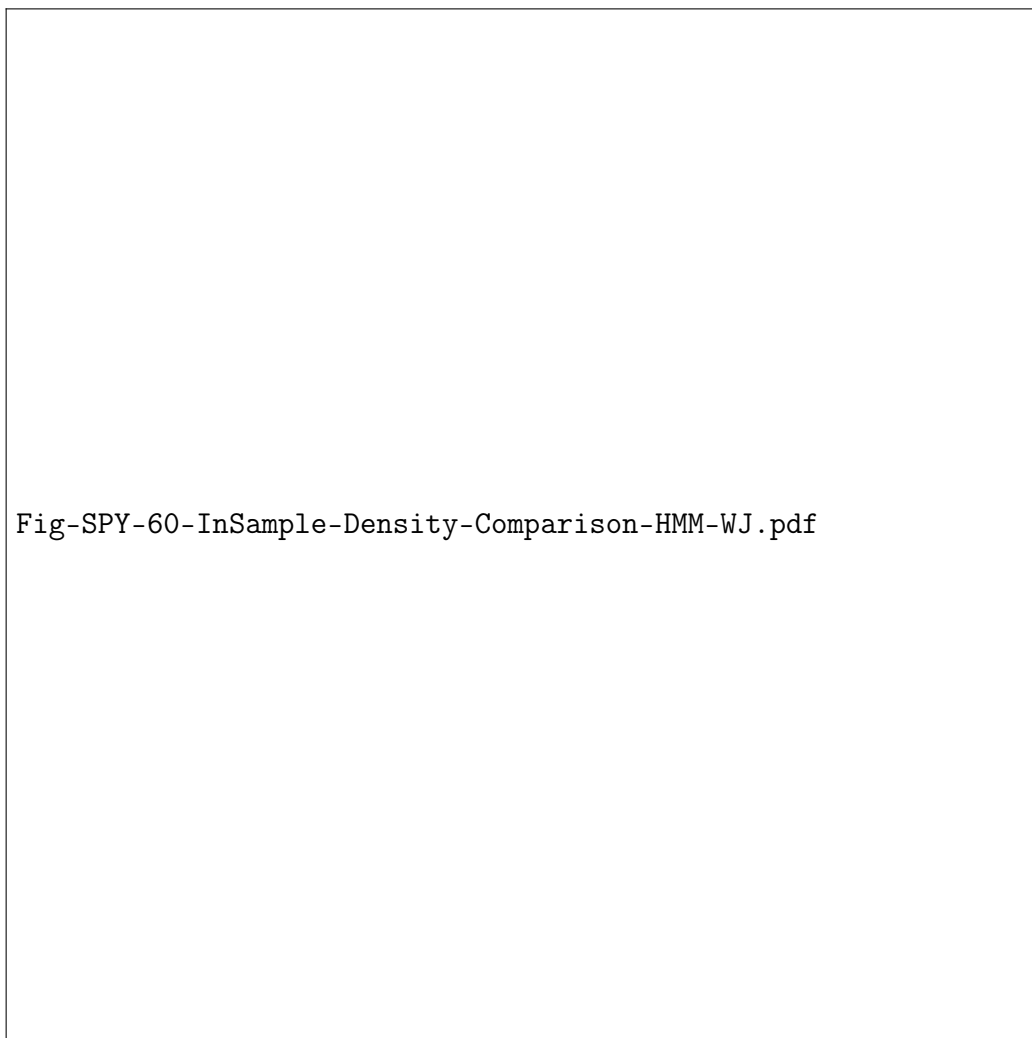
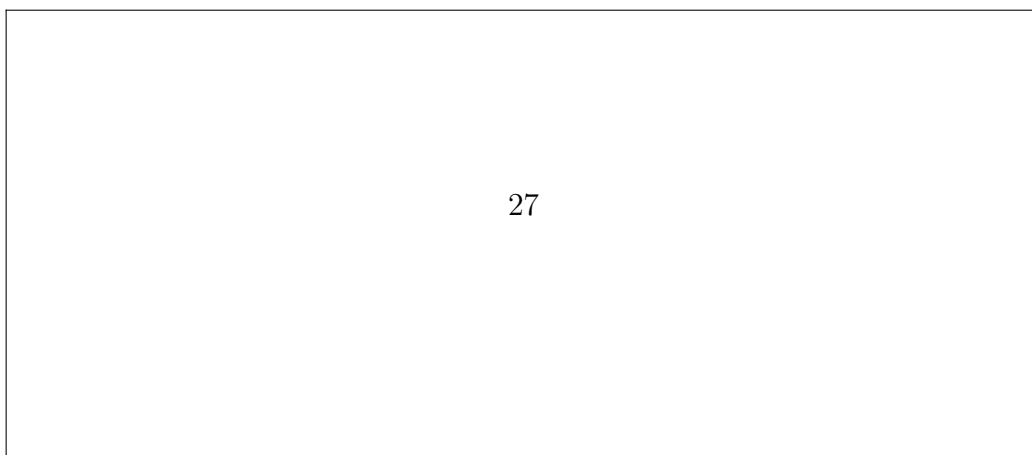


Figure 26: Density (hybrid)

0.48

(a)

b



Algorithm 1 Empirical hidden Markov model construction via Laplace partitioning

Price history $P_{1:T}$, Risk-free rate r_f , Number of states N Transition matrix $\mathbf{T}$, Quantile boundaries $\mathcal{Q}$ Compute excess growth rates $R_t = \frac{1}{\Delta t} \ln(P_t/P_{t-1}) - r_f$ for all t . Fit Laplace parameters (μ_L, b_L) to the series $\mathbf{R}$ using maximum-likelihood estimation. Define interior boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \dots, N-1\}$, where F_L^{-1} is the inverse CDF of the Laplace distribution. Set finite outer bounds $Q_0 = F_L^{-1}(0.001; \mu_L, b_L)$ and $Q_N = F_L^{-1}(0.999; \mu_L, b_L)$. Assign each excess growth rate R_t to a discrete state $s_t \in \{1, \dots, N\}$ based on the boundaries in $\mathcal{Q} = \{Q_k\}$. Initialize count matrix $\mathbf{C} \in \mathbb{R}^{N \times N}$ and transition matrix $\mathbf{T} \in \mathbb{R}^{N \times N}$ with zeros. $t = 1$ to $T-1$ Identify transition $i = s_t$ to $j = s_{t+1}$. $\mathbf{C}_{i,j} \leftarrow \mathbf{C}_{i,j} + 1$. $i = 1$ to N $\mathbf{T}_{i,:} \leftarrow \mathbf{C}_{i,:} / \sum_{j=1}^N \mathbf{C}_{i,j}$ Row-wise normalization $\mathbf{T}, \mathcal{Q}$

Algorithm 2 Hybrid jump-diffusion simulation with persistent volatility

Model parameters $\mathbf{T}, \bar{\pi}, \epsilon, \lambda, N_{tail}, p_{neg}$, Total Steps M Simulated state sequence $S_{1:M}$ Define tail-states $\mathcal{S}_{bottom} = \{1, \dots, N_{tail}\}$ and $\mathcal{S}_{top} = \{N - N_{tail} + 1, \dots, N\}$. Sample initial state $S_1 \sim \text{Categorical}(\bar{\pi})$. Set $counter \leftarrow 2$. $counter \leq M$ Sample $u \sim \text{Uniform}(0, 1)$. $u < \epsilon$ Sample jump-duration $K \sim \text{Poisson}(\lambda)$. $j = 1$ to K while $counter \leq M$ Sample $w \sim \text{Uniform}(0, 1)$. $w < p_{neg}$ $S_{counter} \sim \text{Uniform}(\mathcal{S}_{bottom})$ Bias toward negative-tail events $S_{counter} \sim \text{Uniform}(\mathcal{S}_{top})$ $counter \leftarrow counter + 1$. Sample $S_{counter} \sim \text{Categorical}(\mathbf{T}_{S_{counter-1}, :})$. $counter \leftarrow counter + 1$. $S_{1:M}$

Algorithm 3 State decoding and continuous excess growth rate reconstruction

Simulated state sequence $S_{1:M}$, Quantile boundaries $\mathcal{Q}$ Reconstructed continuous excess growth rates $\hat{R}_{1:M}$ Define $z_{0.999} = 3.09$. $k = 1$ to N Set $\mu_k = (Q_{k-1} + Q_k)/2$. Set $\sigma_k = (Q_k - Q_{k-1})/(2z_{0.999})$. Initialize the reconstructed excess growth rate vector $\hat{\mathbf{R}}$ of length M . $t = 1$ to M Retrieve the simulated state $k = S_t$. Sample the continuous excess growth rate $\hat{R}_t$ from the normal distribution $\mathcal{N}(\mu_k, \sigma_k)$. $\hat{R}_{1:M}$

Algorithm 4 Multi-objective grid search for jump hyperparameters

Observed growth rates $\mathbf{R}_{obs}$, ACF max lag $L = 252$, Number of paths $P = 1000$, Time steps $M = 2766$ Grid space for $\epsilon \in [10^{-5}, 10^{-4}]$ and $\lambda \in [1.0, 100.0]$ Kurtosis penalty weight $w_K = 0.20$ Optimal jump hyperparameters (ϵ^*, λ^*) Calculate observed absolute ACF: $A_{obs}(\tau) = \text{ACF}(|\mathbf{R}_{obs}|, \tau)$ for $\tau \in \{1, \dots, L\}$. Calculate observed global kurtosis: $K_{obs} = \text{Kurtosis}(\mathbf{R}_{obs})$. Initialize minimum error $E_{min} \leftarrow \infty$. each ϵ in grid space each λ in grid space Initialize simulated ACF accumulator $\mathbf{A}_{sum} \leftarrow \mathbf{0}$ and kurtosis accumulator $K_{sum} \leftarrow 0$. $p = 1$ to P Simulate path using Algorithm 2 to obtain state sequence $S_{1:M}$. Decode sequence using Algorithm 3 to obtain continuous rates $\hat{\mathbf{R}}^{(p)}$. $\mathbf{A}_{sum}(\tau) \leftarrow \mathbf{A}_{sum}(\tau) + \text{ACF}(|\hat{\mathbf{R}}^{(p)}|, \tau)$ for all τ . $K_{sum} \leftarrow K_{sum} + \text{Kurtosis}(\hat{\mathbf{R}}^{(p)})$. Compute ensemble averages: $\bar{A}_{sim}(\tau) = \mathbf{A}_{sum}(\tau)/P$ and $\bar{K}_{sim} = K_{sum}/P$. Compute objective: $J = \sum_{\tau=1}^L (A_{obs}(\tau) - \bar{A}_{sim}(\tau))^2 + w_K (K_{obs} - \bar{K}_{sim})^2$. $J < E_{min}$ $E_{min} \leftarrow J$ $(\epsilon^*, \lambda^*) \leftarrow (\epsilon, \lambda)$ ϵ^*, λ^*

Supplemental Information